

Single-Event Upset Resilience Utilizing Hyperdimensional Computing for Aerospace Prone Computer Systems Studied With GOES-16 Satellite X-ray Flux Data

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Abstract—This paper presents a method to maintain classification performance despite distorted data caused by Single-Event Upset (SEU) bit flips in aerospace systems onboard volatile and non-volatile memory via Hyperdimensional Computing (HDC). Classification accuracy is evaluated across two trials using solar flare detection on the GOES-16 X-ray flux dataset. In the first trial, bit flips are injected transiently during model inference at rates from 0–30%, simulating single-event upsets in volatile RAM. In the second trial, bit flips accumulate directly in model weights over a simulated 10-year mission lifetime across three solar cycle profiles, affecting both volatile and non-volatile memory. HDC is evaluated against random forest, support vector machine, and neural network baselines. Results show HDC maintains 91.2% accuracy at 30% corruption compared to 48.2–53.7% for traditional ML baselines, representing up to a 1.9x robustness advantage in trial 1. Under simulated 10-year mission profiles, traditional ML models degraded to near-random classification within 1–2 years while HDC maintained 68.8–81.3% accuracy across all three solar cycle scenarios as seen in trial 2. These findings are significant for aerospace systems that are particularly prone to radiation-induced memory errors. Applications include but are not limited to Attitude Determination and Control Subsystem (ADCS) star trackers, reaction wheel algorithms, and magnetometer algorithms. Solar flare detection, Earth weather pattern classification, and fault detection, isolation, and recovery (FDIR) systems too.

I. INTRODUCTION

Electronic aerospace systems face a unique computational challenge regarding Single-Event Upset (SEU) related bit flips in memory. Vehicles found in low Earth orbit (LEO), like the International Space Station (ISS), Spy Satellites, Star Link, and Cube Satellites (CubeSats) all experience these unwanted bit flips [1], [2]. As electronic systems move further away from Earth's magnetosphere, they are less shielded from the harsh radiation environment that is space. This becomes a more critical and frequent issue for medium Earth orbit (MEO), geostationary Earth orbit (GEO), and deep space vehicles like the James Webb and Hubble telescope as they are exposed to the Van Allen Belts, galactic cosmic rays, and solar particle events respectively [3].

To overcome these challenges, we propose a Hyperdimensional computing (HDC) based software and hardware architecture that is both computationally and memory-efficient. HDC is a brain-inspired computing paradigm that represents information using high-dimensional vectors (hypervectors-HVs) with thousands of bipolar elements [4]. HVs are manipulated through basic operations such as *binding*, *bundling*, and *permutation* to encode symbolic and structural information. Unlike conventional ML, HDC-based systems are resilient to noise and well-suited for hardware implementation as they rely on simple arithmetics rather than computation-intensive floating-point operations [5].

In our research, we simulate radiation-induced bit flips affecting onboard volatile memory at incremental intensities, which is then used to classify solar flare detections using the GOES-16 satellite X-ray flux dataset [6]. Specifically, in our first trial, we insert bit flips at a specified rate during model inference as a transient one-time corruption that disappears after each prediction. The stored class prototypes stay clean, and our models are able to show how they handle one-time bit flips at different intensities. In our second trial, we insert bit flips at a specified rate per orbit to the model weights over time. This simulates bit flips occurring over time in volatile and non-volatile memory and the effects we can see on its classification abilities as SEU bit flips occurs. We utilize neural networks, support vector machine (SVM), random forest, and

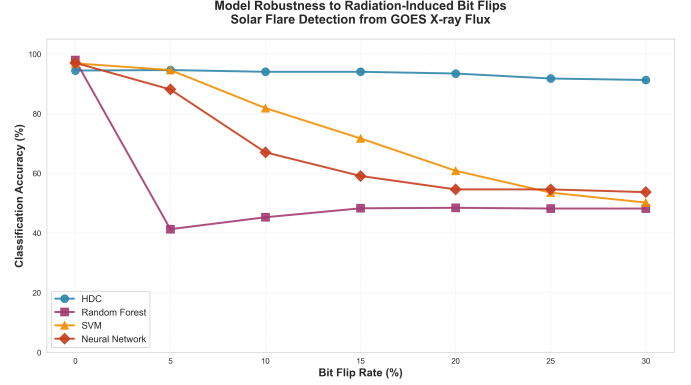


Fig. 1. Classification accuracy of HDC, Random Forest, SVM, and Neural Network models under increasing bit flip rates from 0–30%, simulating single-event upsets in volatile RAM during inference. HDC maintains 91.2% accuracy at 30% corruption while traditional ML baselines degrade to 48.2–53.7%.

HDC classification models as our primary candidates in this study.

In efforts to produce accurate findings, as illustrated in Fig. 2, this study takes into account the GOES-16 satellite X-ray flux dataset's satellite orbit and the realistic radiation exposure ranges within that orbit. The assumptions made are that GOES-16 operates in geostationary orbit with a 24-hour orbital period, meaning one orbit equals one day of radiation exposure. Bit flips are modeled based on published GEO commercial SEU rates of approximately 10^{-5} errors/bit/day [1], serving as a baseline rate during periods of low solar activity. In the first trial, bit flips are applied transiently to hypervectors held in volatile RAM during active inference, representing SEU's that occur and resolve within one prediction cycle. In the second trial, bit flips are applied cumulatively to model weights over time, representing SEU damage accumulating in both volatile RAM during operation and non-volatile memory where weights persist across power cycles. For HDC, this corruption targets the class prototype hypervectors directly, each consisting of 10,000 bits (1.2 KB). For the neural network, SVM, and random forest models, equivalent corruption is applied as scaled input noise proportional to the cumulative flip fraction, as these models do not expose inspectable prototype memory. No Error Correction Code (ECC) memory scrubbing is assumed for both trials, representing a worst-case Commercial Off-The-Shelf (COTS) hardware deployment scenario [7], [8]. To model realistic long-duration mission exposure, SEU rates are varied across solar cycle phases [9] over a simulated 10-year mission lifetime: solar minimum (1 flip/orbit), ascending phase (3 flips/orbit), solar maximum (5 flips/orbit), descending phase (2 flips/orbit), and solar particle events (50 flips/orbit). Three mission profiles are evaluated based on the solar cycle phase at launch: launch at solar minimum, launch at solar maximum, and an active cycle scenario with Solar Proton Events (SPE) occurring during solar maximum. Model accuracy is sampled monthly (every 30 orbits) across all three profiles.

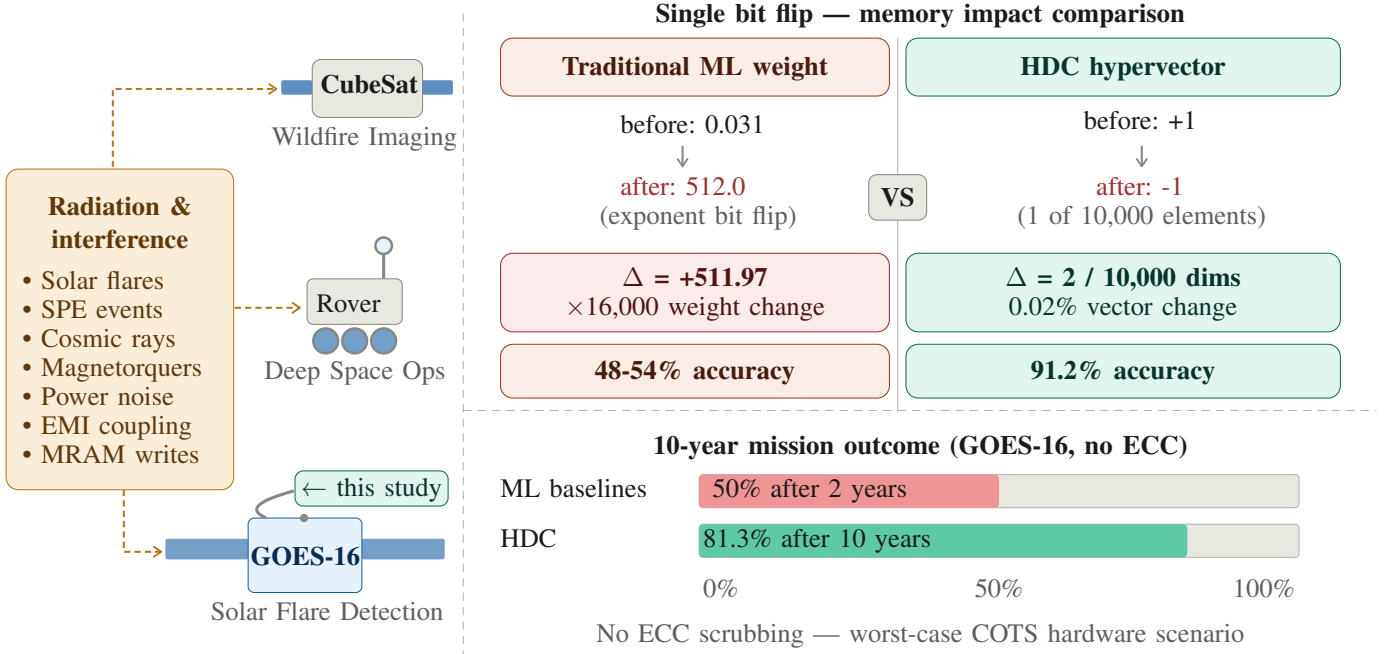


Fig. 2. Radiation and interference sources affecting onboard classification systems across spacecraft platforms. A single bit flip in a traditional ML weight causes a $\times 16,000$ magnitude change, while the same flip in an HDC hypervector changes only 1 of 10,000 elements (0.02%). Under a simulated 10-year GOES-16 GEO mission with no ECC scrubbing, ML baselines degrade to $\sim 50\%$ accuracy within 1–2 years while HDC maintains 81.3% at year 10.

II. BACKGROUND AND LITERATURE REVIEW

A. Radiation and Memory Corruption in Aerospace Systems

Electronic systems in orbit face a constant bombardment of charged particles. Galactic cosmic rays arrive from outside the solar system at all times, trapped protons and electrons concentrate in the Van Allen Belts at MEO altitudes, and solar particle events release intense bursts of heavy ions during periods of high solar activity [2], [3]. Any of these particles can deposit enough charge in a semiconductor junction to flip a stored bit, a phenomenon called a single-event upset (SEU) [1]. In volatile SRAM, the flipped bit persists only until the cell is rewritten. In non-volatile charge-based memories like Flash and EEPROM, the damage can be worse. For example, cumulative total ionizing dose (TID) degrades gate oxide over months and years of exposure, and a single heavy ion strike can corrupt stored data that survives power cycles [3]. For commercial hardware with no radiation controls, the published SEU rate is approximately 10^{-5} errors/bit/day [1]. Rad-tolerant parts improve on this by two to three orders of magnitude at 10^{-7} to 10^{-8} , and fully radiation-hardened components reach 10^{-10} to 10^{-12} [1].

Radiation accounts for the majority of memory corruption events in space, but it is not the only contributor. Thermal cycling between direct sunlight and eclipse shadow shifts transistor threshold voltages, and at the extremes of deep-space temperature ranges these shifts can produce read errors that look identical to bit flips at the software level. On densely packed CubeSat and PocketSat platforms, internal electromagnetic interference from power converters, motor drivers, and radio transmitters couples into memory buses with minimal physical separation to act as a buffer. Voltage glitches during thruster firing, eclipse entry, or battery switchover can latch incorrect values into memory cells as well.

Single-event latch-up (SEL) presents a more severe failure mode. A heavy ion can trigger a parasitic thyristor structure inherent to CMOS bulk devices [1]. The device locks into a high-current state and holds corrupted data until the entire subsystem is power-cycled. If the latch-up goes undetected for even a few microseconds, the excessive current can cause permanent damage [1]. Radiation-hardened CMOS-SOI and CMOS-SOS fabrication processes eliminate the parasitic

PNPN path responsible for this failure mode, but COTS hardware carries no such protection.

From the perspective of onboard classification software, all of these mechanisms are indistinguishable. A corrupted bit is a corrupted bit regardless of whether radiation, a thermal excursion, or a power glitch caused it. COTS hardware represents the worst case for any of these failure modes because it has no radiation hardening, no ECC scrubbing, and no latch-up immunity [1], [2]. We adopt this worst-case COTS assumption throughout our study.

B. Hyperdimensional Computing (HDC)

Hyperdimensional Computing (HDC) is a brain-inspired computing paradigm first formalized by Kanerva [10] that represents all data as high-dimensional vectors called hypervectors (HVs). A hypervector is a vector with thousands of elements drawn from a bipolar alphabet of +1 and -1. In our implementation, each hypervector contains 10,000 such elements. Two randomly generated hypervectors of that size will be nearly orthogonal to each other with overwhelming probability, which gives the system a vast capacity to represent distinct concepts without interference [10].

HDC operates through three core vector operations. Binding associates two hypervectors through element-wise multiplication, producing a new hypervector that is dissimilar to both inputs and encodes their paired relationship. Bundling combines multiple hypervectors through element-wise addition followed by a sign function, producing a composite that is similar to all its inputs. We use bundling to accumulate all bound pairs in a data window into a single encoded hypervector. Permutation shifts the elements of a hypervector by a fixed number of positions, encoding sequential order when temporal structure matters.

Training in HDC requires only a single pass through the data. Each training sample is encoded into a hypervector using the operations above, and all hypervectors belonging to the same class are bundled together into a single class prototype. No gradient computation, no backpropagation, and no iterative weight tuning are involved. A new sample is classified by encoding it and measuring its cosine similarity against each stored class prototype. The class with the highest similarity wins. The radiation tolerance argument for HDC follows directly from this representation. Every element in

a bipolar hypervector carries equal informational weight. No single dimension is more important than any other. A SEU bit flip in memory that corrupts one element of a 10,000-dimensional hypervector changes a $+1$ to a -1 or vice versa. That flip alters 2 out of 10,000 dimensions, a 0.02% change in the vector. The cosine similarity between the original and corrupted vector shifts by an amount so small that classification decisions remain intact. Barkam demonstrated that HDC models retain up to 90% higher accuracy than deep neural networks when subject to random noise injected into model weights, confirming that the distributed nature of hyperdimensional representations provides intrinsic fault tolerance without any additional error correction [5]. The equal weight property of bipolar elements is what separates HDC from architectures that store parameters in floating point, where a single bit flip can produce catastrophic changes to a weight value.

C. Traditional ML in Radiation Environments

Random forest, support vector machine, and neural network classifiers all store their learned parameters as IEEE 754 floating-point values. A 32-bit float encodes a number using one sign bit, eight exponent bits, and twenty-three mantissa bits. The exponent bits carry disproportionate influence over the magnitude of the stored value. If a single bit flip strikes one of those eight exponent bits, the represented number can jump by powers of two. A weight originally stored as 0.031 can become 512.0 after a single exponent bit corruption, a change of approximately $\times 16,000$ in magnitude as illustrated in Fig. 2. A flip in the sign bit reverses the polarity of the weight entirely. Only a flip in the lower mantissa bits produces a small, tolerable perturbation, and those bits occupy less than three quarters of the 32-bit word.

Random forest stores hundreds of split thresholds and leaf values across its decision trees, each as a floating-point number. A corrupted split threshold reroutes entire branches of the tree, sending test samples down incorrect paths. SVM stores support vectors and dual coefficients that define the decision boundary in kernel space. A single corrupted coefficient can shift or collapse that boundary with no warning to the inference pipeline. Neural networks store weight matrices across every layer, and a corrupted weight in an early layer propagates its error through all subsequent layers during forward computation. The architectures differ in how they learn, but they share the same storage format and the same vulnerability: every parameter sits in floating-point memory where a single bit flip in the wrong position can change the value by orders of magnitude.

HDC stores its class prototypes as bipolar vectors of $+1$ and -1 values. A bit flip in one element changes that element's contribution by exactly 2 out of 10,000 dimensions. No element is more important than any other, and no single corruption event can produce the catastrophic magnitude shifts that floating-point representations allow [5]. We quantify this contrast directly in our experimental results.

D. Research Gap and Motivation

Prior work has demonstrated that HDC tolerates noise in model weights better than deep neural networks under controlled perturbation scenarios [5]. Separate studies have applied HDC to spacecraft telemetry anomaly detection [7] and proposed bitstream resilience layers for radiation-exposed FPGA configurations [8]. What has not been done is a direct comparison of HDC against traditional ML baselines under realistic mission-lifetime SEU accumulation using real space weather data as the classification target. Our work fills that gap.

Simulated bit flip injection is the standard methodology in this field. No publicly available satellite downlink dataset contains uncorrected bit errors because onboard ECC and ground-station Reed-Solomon decoding remove them before archiving. Every published study that evaluates HDC or ML robustness to memory corruption relies on artificial injection of bit flips into model memory during or after training [7], [8]. Our approach follows the same convention and is directly comparable to existing results.

Several operational scenarios motivated this study. Star tracker classifiers that identify constellation patterns for attitude determination must produce correct outputs even after



Fig. 3. Model accuracy at 0% and 30% bit flip rates. HDC sustains near-baseline performance while Random Forest, SVM, and Neural Network baselines drop to near-random classification.

years of radiation exposure, because a misidentified star field feeds a wrong quaternion to the attitude control loop. Reaction wheel control algorithms in the ADCS depend on stored weights to compute torque commands and a corrupted weight can produce an uncontrolled torque output that tumbles the spacecraft. On deep-space rovers and long-duration probes, ground operators cannot intervene in real time, so the onboard classifier must degrade gracefully over years without a reboot option. CubeSat missions performing onboard image triage for wildfire detection or Earth observation face tight downlink bandwidth constraints that make onboard classification essential, and corruption tolerance protects the triage decision from silent memory errors that accumulate between ground contacts.

We evaluate these robustness questions using the GOES-16 satellite X-ray flux dataset as our experimental platform [6]. Solar flare detection from X-ray time series provides a well-defined binary classification task with real operational relevance to space weather monitoring, and the GOES-16 geostationary orbit places the sensor in an environment where the radiation threats described in Section II are present at every moment of the mission.

III. METHODOLOGY

A. Dataset

We use the GOES-16 X-Ray Sensor A (XRSA) channel covering the 1–8 Å wavelength band, downloaded via SunPy from NOAA archives for the period of January through June 2023 [6]. The raw time series contains approximately 15.7 million one-second cadence samples. We detect flares algorithmically by identifying contiguous intervals where the X-ray flux exceeds the NOAA C-class threshold of 10^{-6} W/m², requiring a minimum duration of five minutes and classifying each event by its peak flux against standard NOAA thresholds. This procedure identifies 1,847 flare events across C, M, and X classes. For binary classification, any flare detection within a 60-minute prediction horizon following a 120-sample input window counts as a positive label. The resulting dataset is heavily imbalanced toward the non-flare class. We balance it to 3,000 total samples (1,500 per class) using undersampling of the majority class and noise-augmented oversampling of the minority class, then split 70/30 into training and test sets with stratified sampling.

B. HDC Implementation

Each 120-sample input window is encoded into a 10,000-dimensional bipolar hypervector. The encoder first applies a log-scale quantization of the raw flux values into 100 bins. Each bin maps to a pre-generated level hypervector in $\{-1, +1\}^{10000}$, and each of the 120 time positions maps to a separate position hypervector of the same size. For a given

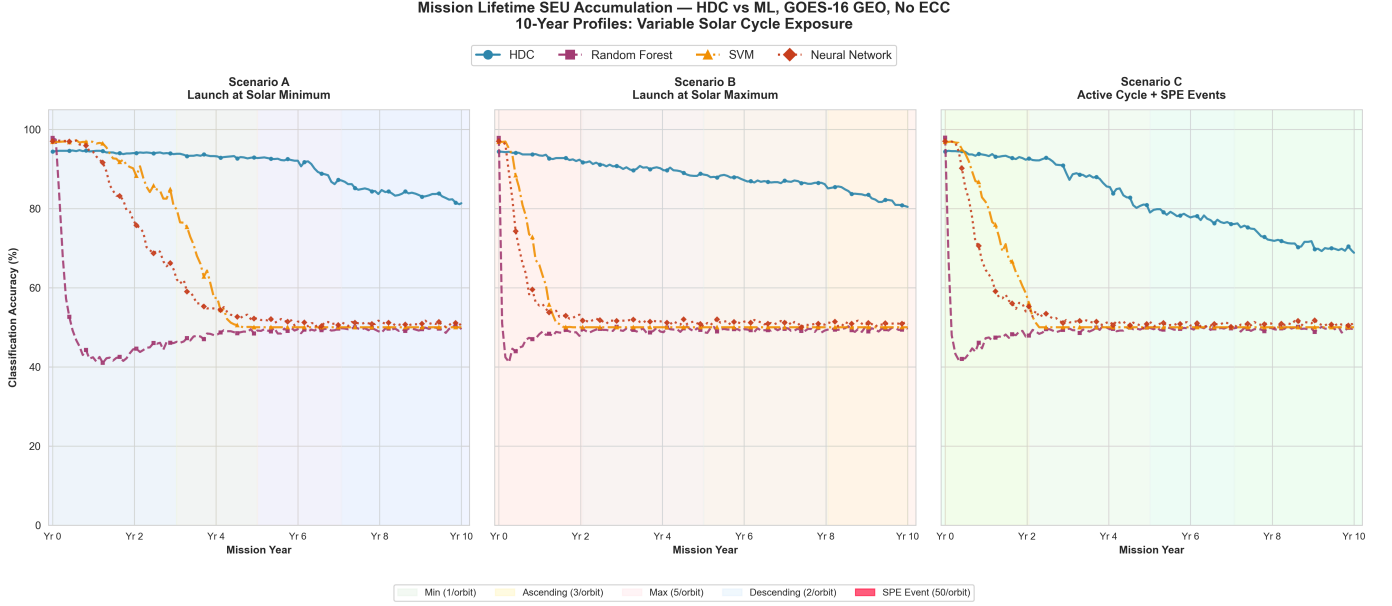


Fig. 4. 10-year cumulative SEU degradation across three mission profiles: launch at solar minimum (A), launch at solar maximum (B), and active cycle with SPE events (C). Traditional ML models degrade to near-random classification within 1–2 years across all scenarios. HDC maintains 68.8–81.3% accuracy at year 10 depending on solar exposure.

time step, the level hypervector is bound to the position hypervector through element-wise multiplication. All 120 bound vectors are then bundled by element-wise summation followed by a sign function, producing a single encoded hypervector for the window. During training, all encoded hypervectors belonging to a given class are bundled together into one class prototype. Two prototypes result from our binary task. Inference encodes a test window using the same pipeline and classifies it by cosine similarity against both prototypes. The class with the higher similarity score is the prediction. The entire training process requires a single pass through the data with no iterative optimization.

C. ML Baselines

We train three traditional ML classifiers on the same 70/30 split. Random forest uses 100 estimators with a maximum depth of 20. SVM uses a radial basis function kernel with $C = 10.0$ and gamma set to scale. The neural network is a multi-layer perceptron with two hidden layers of 100 and 50 neurons, trained for up to 500 iterations with early stopping enabled. All three models receive standardized input features. Each model is seeded with random state 42 for reproducibility.

D. Trial 1: Transient Bit Flip Injection

In the first trial, we inject bit flips transiently into the test hypervectors during inference at rates from 0% to 30% in 5% increments. At each rate, a fraction of the 10,000 dimensions in the encoded test hypervector are selected uniformly at random and multiplied by -1 , flipping them from $+1$ to -1 or vice versa. The stored class prototypes remain clean. Each corrupted test hypervector is then compared against the uncorrupted prototypes using cosine similarity. For the ML baselines, equivalent corruption is applied as scaled Gaussian noise on the standardized input features, with noise magnitude proportional to the bit flip rate. This trial isolates how each model handles one-time corruption events that occur and resolve within a single prediction cycle, representative of transient SEUs in volatile RAM during active inference.

E. Trial 2: Cumulative Mission-Lifetime SEU Accumulation

The second trial simulates 10 years of radiation exposure on a GOES-16 geostationary orbit with a 24-hour orbital period,

giving 3,650 total orbits [9]. We assume a baseline GEO commercial SEU rate of approximately 10^{-5} errors/bit/day applied to the 10,000-bit HDC class prototypes [1]. SEU rates vary by solar cycle phase [3]: solar minimum at 1 flip/orbit, ascending phase at 3 flips/orbit, solar maximum at 5 flips/orbit, descending phase at 2 flips/orbit, and solar particle events at 50 flips/orbit. On each orbit, the specified number of bits are selected uniformly at random from the live class prototypes and flipped. Bits that were previously flipped can flip back, which is realistic SEU behavior. For the ML baselines, equivalent corruption is applied as cumulative Gaussian noise scaled to the accumulated flip fraction.

Three mission profiles define the solar cycle phase sequence over the 10-year window. Profile A launches at solar minimum and follows the standard 11-year cycle ordering: 3 years minimum, 2 years ascending, 2 years maximum, 3 years descending. Profile B launches at solar maximum with the same phase durations rotated. Profile C starts in the ascending phase with 3 years of solar maximum and includes 10 randomly placed SPE bursts of 3 orbits each during the maximum phase, making it the harshest scenario [9], [3]. Accuracy is checkpointed every 30 orbits (approximately monthly) across all profiles. No ECC memory scrubbing is assumed for either trial, representing a worst-case COTS hardware deployment [1], [2].

IV. RESULTS

A. Trial 1: Transient Bit Flip Robustness

Fig. 1 plots classification accuracy against bit flip rate for all four models from 0% to 30%. At 0% corruption, HDC achieves 94.4% accuracy. Random forest leads at 97.9%, SVM reaches 96.8%, and the neural network reaches 97.0%. All three ML baselines outperform HDC on clean data, which is expected given that floating-point models have greater representational precision than bipolar vectors for uncorrupted inputs.

The gap reverses as corruption increases. Random forest collapses first, dropping from 97.9% to 41.3% at just 5% bit flips and never recovering, settling at 48.2% by 30%. The neural network degrades more gradually, falling through 88.1% at 5%, 67.0% at 10%, and reaching 53.7% at 30%. SVM follows a similar trajectory, holding at 94.6% through 5% but then dropping to 81.8% at 10% and 50.2% at 30%. All three ML models converge near 50% at the highest corruption rate, which is equivalent to random guessing on a balanced binary task.

HDC tells a different story. Accuracy at 5% corruption is 94.6%, essentially unchanged from the clean baseline. At 10% it holds at 94.0%. Even at 30% corruption, HDC maintains 91.2% accuracy, a relative degradation of only 3.4% from its clean performance. As shown in Fig. 3, the best ML baseline at 30% is the neural network at 53.7%, giving HDC a $1.9\times$ robustness advantage measured as the ratio of degraded accuracies. The ML models lose between 44.7% and 50.8% of their clean accuracy at 30% corruption. HDC loses 3.4%. The operational meaning of this gap is straightforward: an HDC classifier deployed on COTS hardware would continue producing usable predictions under corruption levels that render every traditional ML baseline no better than a coin flip.

B. Trial 2: Mission-Lifetime SEU Accumulation

Fig. 4 shows accuracy over 10 years for all four models across the three mission profiles. Random forest is the first to fail in every scenario. Under Profile A (launch at solar minimum), random forest drops below 60% by year 0.33, before the spacecraft has completed its first four months of operation. The neural network follows at year 3.29 and SVM at year 3.95. Under Profile B (launch at solar maximum), the higher initial SEU rate accelerates every collapse: random forest fails by year 0.08, the neural network by year 0.74, and SVM by year 1.23. Profile C is the harshest scenario because its 3-year solar maximum window includes 10 SPE bursts at 50 flips/orbit. Random forest fails by year 0.16, the neural network by year 1.23, and SVM by year 1.97. Across all three profiles, every ML model converges to approximately 50% accuracy and stays there for the remainder of the mission.

HDC degrades but never collapses. At year 10 under Profile A, HDC holds 81.3% accuracy. Profile B produces a similar result at 80.4%. Profile C, with its concentrated radiation exposure, pushes HDC down to 68.8% at year 10. Even in this worst case, HDC remains nearly 19 percentage points above random guessing and is the only model still producing operationally useful classifications. The ML baselines have been indistinguishable from a coin flip for eight or more years at that point.

Profile C deserves particular attention because it represents the most operationally relevant threat. SPE events are unpredictable and can deliver radiation doses in hours that would otherwise accumulate over weeks of normal solar maximum exposure [3]. A classifier that fails under SPE conditions fails exactly when the spacecraft needs it most, during the period of highest radiation activity when accurate solar flare detection has the greatest operational value. HDC is the only model in our evaluation that survives these conditions with meaningful classification accuracy intact at mission end.

V. CONCLUSION

We evaluated HDC against random forest, SVM, and neural network classifiers under two radiation-induced bit flip scenarios using the GOES-16 X-ray flux dataset for solar flare detection. In Trial 1, transient corruption at 30% of model memory reduced the ML baselines to 48.2–53.7% accuracy while HDC held at 91.2%, a $1.9\times$ robustness advantage. In Trial 2, cumulative SEU accumulation over a simulated 10-year GEO mission drove every ML model to near-random classification within one to two years. HDC maintained 68.8–81.3% accuracy at year 10 across all three solar cycle profiles, including the harshest scenario with SPE events during solar maximum.

The mission-lifetime result carries a practical implication for COTS-based spacecraft design. An HDC classifier deployed without ECC scrubbing or radiation hardening remains the only model in our evaluation that produces operationally useful outputs after a full decade of unprotected exposure. The three ML baselines become useless early enough in the mission that their initial accuracy advantage over HDC on clean data is irrelevant for any deployment lasting more than two years.

The robustness we measured is not specific to radiation. Because all corruption sources described in Section II produce the same effect at the software level, the bit flip tolerance

demonstrated here applies equally to thermal errors, EMI-induced corruption, and power transient glitches. Any environment where memory contents change without warning is an environment where HDC outperforms floating-point ML.

Future work will extend this study in three directions. First, we plan to evaluate the effect of periodic ECC scrubbing at various intervals to determine how much correction is needed before traditional ML models become competitive with unscrubbed HDC. Second, we intend to implement the HDC encoder and classifier on FPGA hardware to measure inference latency and power consumption under real-time constraints. Third, we will extend the binary classification task to multi-class solar flare severity prediction across C, M, and X classes to evaluate whether HDC robustness scales to finer-grained decision boundaries.

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